Computer Science & IT



Data Structure & Programming

Array

Lecture No. 02



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Recap of Previous Lecture



- Topic
- Topic
- Topic
- Topic
- Topic

1-D array

2-D array

Row major order

No. of Rows / # columns

Completed

(i - LB)

Total No. of Rows / total No. of column

(UB - LB + 1)

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Generalized formula Row major

Column major order

Lower Triangular Matrix
- RMO
- CMO

Slide



Question : 1-D Array

Consider the above array A[-5.....100]. Base address of the array is 1000 and each element occupies 2-Bytes of space. What is the address of A[70]?

(A) 1150

(B) 1250

(C) 1200

(D) 1350

$$BA + (i - LB) \times Size$$

$$1000 + (70 - (-5)) \times 2$$

$$1000 + 75 \times 2 = 1150$$

Slide



Question : 2-D Array, MSQ

Consider the following array A[-23…93] [-7…31].Base address is 1000 and each element occupies 2-Bytes of space. Which of the following is TRUE ?

117 ← LB_1 UB_1

LB_2 ← UB_2 → $31-(-7)+1$

39 column

(A) The number of rows is 117 ~~elements in a row~~ → $UB_1 - LB_1 + 1 = 93-(-23)+1$

$93+23+1$

117

(B) The Array B[0..116] [2..40] have same number of elements as the array declared in question

$116-0+1=117$.

$40-2+1=39$

(C) Address of A[0][5] is 3538 row major order ✗

(D) Address of A[0][0] is 3538 row major order ✗

Slide



Question : 2-D Array, MSQ

Consider the following array A[-23...93] [-7...31].Base address is 1000 and each element occupies 2-Bytes of space. Which of the following is TRUE ?

total = $39 \times 23 = 897$ (11)

(C) Address of A[0][5] is 3538 row major order

we are are 0th Row

No. of Rows :- $(0 - (-23)) = 23$ completed

No. of element in each

$31 - (-7) + 1 = 39$

in 0th row

No. of element completed

$(5 - (-7)) = 12$

$1000 + 909 \times 2 = 1000 + 1818$

2818

Slide



Question : 2-D Array, MSQ

Consider the following array A[-23…93] [-7…31].Base address is 1000 and each element occupies 2-Bytes of space. Which of the following is TRUE ?

(D) Address of A[0][0] is 3538 column major order



Topic : 2-D Generalization Row major order:

$A[LB_1 \ldots uB_1][LB_2 \ldots uB_2]$

$A[i][j]$

BA - Base Address

S - Size

$BA + [(i - LB_1) \times (uB_2 - LB_2 + 1) + (j - LB_2)] \times S$

No. of Rows completly arranged before ith Row

No of element in each Row

in ith Row No. of element completed.

Slide

Topic : 2-D Column Major Order

	col-1	col-2	col-3
Row-1	a_{11}	a_{12}	a_{13}
Row-2	a_{21}	a_{22}	a_{23}
Row 3	a_{31}	a_{32}	a_{33}

Before 3rd column, No. of column completed

$(3-1) = 2$

No. of element in each column 3 total $2 \times 3 = 6$

col-1			col-2			col-3		
a_{11}	a_{21}	a_{31}	a_{12}	a_{22}	a_{32}	a_{13}	a_{23}	a_{33}
1000	1002	1004	1006	1008	1010	1012	1014	1016

Address a_{23} — 2nd Row, 3rd column

In 3rd column we are in 2nd Row

No. of element arranged : $2 - 1 = 1$

total $6 + 1 = 7$

Address $a_{23} = 1000 + 7 \times 2 = 1014$

Slide



Topic : 2-D Column Major Order

A [1..100] [1..100]

BA - 100

Size - 2B

Address A[50][49]

CMO

$a_{50,49}$

49th column

No. of column arranged (49-1) : 48

No. of element each column - 100

total = 48 × 100 = 4800

In 49th we are in 50th Row

No. of elements arranged : 50-1 = 49

100 + (4800 + 49) × 2

100 + 4849 × 2 = 9698

100

9798

Slide



Topic : 2-D Column Major Order

Consider the following array A[-4...100] [3...100]. Base address is 100 and each element occupies 2-Bytes of space. What is the address of A[50][48] in column major order ?

$100 + [(48-3) \times (100-(-4)+1) + (50-(-4))] \times 2$

$100 + [45 \times 105 + 54] \times 2$

$4725 + 54 = 4779 \times 2$

9558

$100 + 9558 = 9658$

Slide

$A[LB_1 \ldots UB_1][LB_2 \ldots UB_2]$

Address $A[i][j]$ - column major

Size of column

$$BA + [(j - LB_2) \times (UB_1 - LB_1 + 1) + (i - Lb_1)] \times Size$$



Topic: Sparse Matrix

Sparse Matrix

Sparse matrix relatively few entries are Non zero.

Slide



Topic: Lower Triagular Matrix

Square matrix Raw Size = column Size

Element abave principal diagonal are zero

DPP. PYQ

a_{11}	a_{12} O	a_{13} O	a_{14} O
a_{21}	a_{22}	a_{23} O	a_{24} O
a_{31}	a_{32}	a_{33}	a_{34} O
a_{41}	a_{42}	a_{43}	a_{44}

a_{ij} $j > i$ $a_{ij} = 0$

No. of Nonzero elements $n \times n$

$$1+2+3+4+\dots+n = \frac{n(n+1)}{2}$$

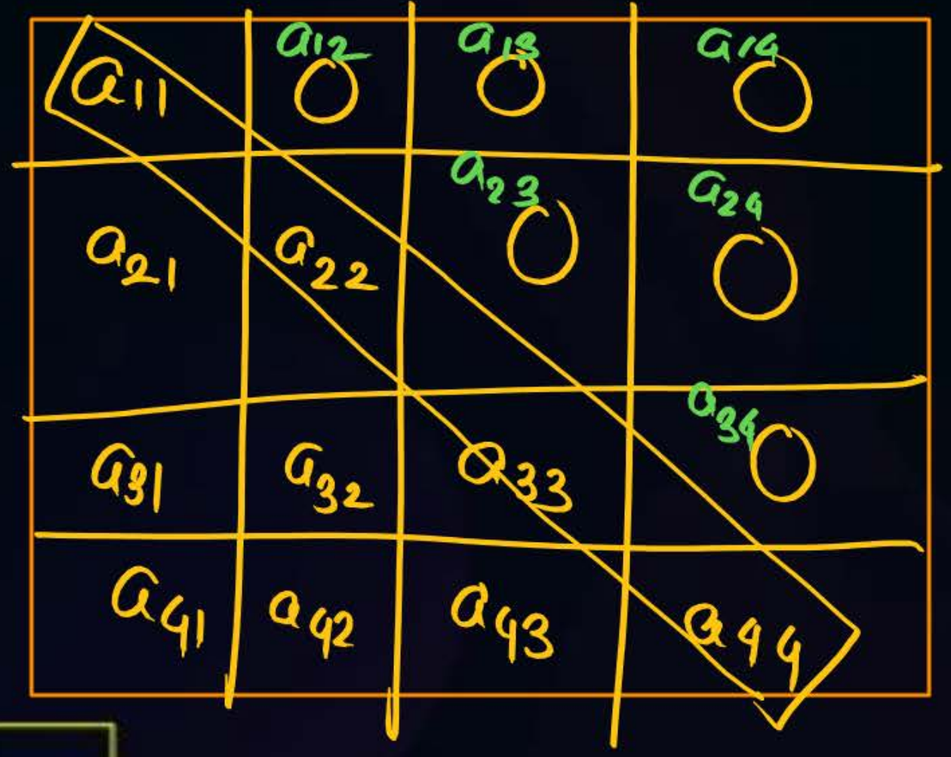
$$\frac{n}{2}(1+n) = \frac{n(n+1)}{2}$$

Slide

Topic: Lower Triagular Matrix

elements above principal diagonal Not store

Row major order :-



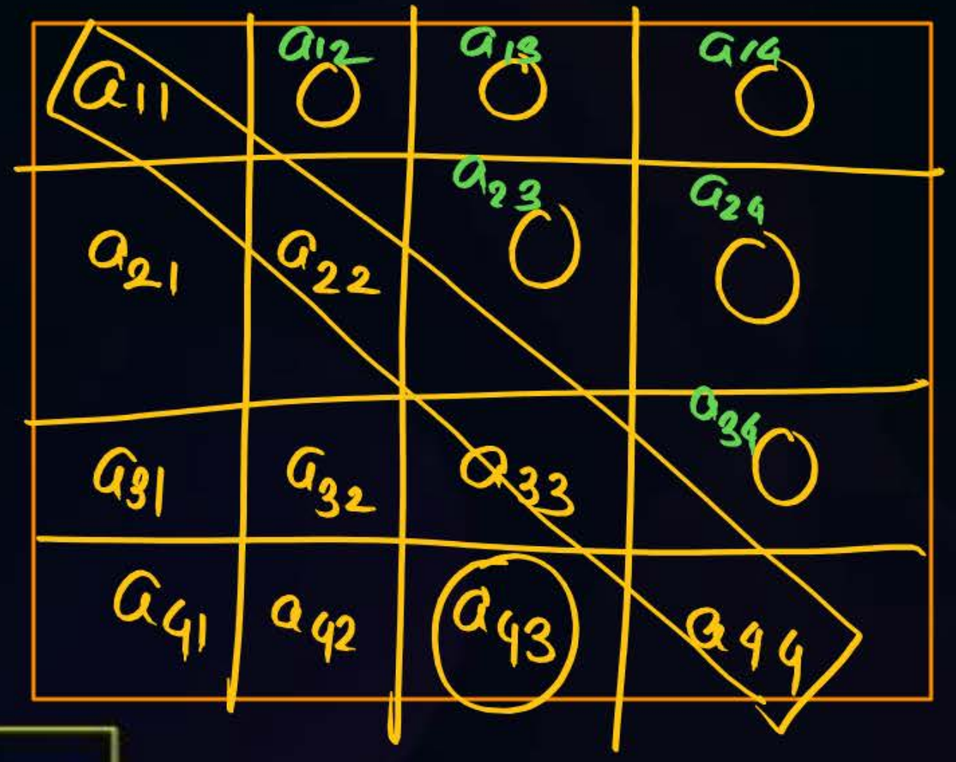
a_{11}	a_{21}	a_{22}	a_{31}	a_{32}	a_{33}	a_{41}	a_{42}	a_{43}	a_{44}
100	102	104	106	108	110	112	114	116	116

1000

a_{43} Address.

Slide

Topic: Lower Triagular Matrix



a_{43}

4th Row 3rd column

No. of Rows completed $4-1=3$ ✓

$n = i-1$

$\frac{(i-1)\times i}{2}$

No. of elements completed 3 Rows

$1+2+3 = \frac{3\times 4}{2} = 6$ ✓

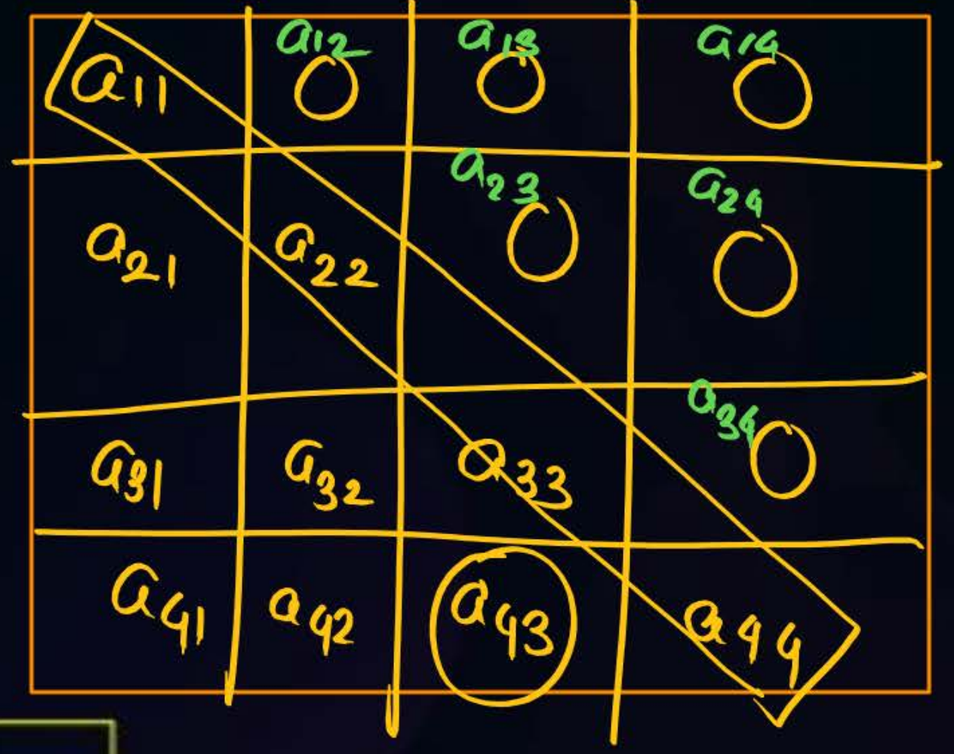
$\frac{n(n+1)}{2}$!!

a_{ij} No. of Row Completed $(i-1)$

No. of element : $1+2+3+\ldots+(i-1) = \frac{(i-1)\,i}{2}$

Slide

Topic: Lower Triagular Matrix



a_{43}

4th Raw 3rd column

$\frac{n(n+1)}{2}$!!

n = ?

In 4th Raw 3rd column

No. of completed $(3-1)=2$

---

a_{ij} ith Raw jth column

in ith Raw No. of element completed

$(j-1)$

total = 6+2 = 8

Address = 100+8x2 = 116

Slide



Topic: Lower Triagular Matrix

A[1...n][1..n]

LTM RMO

$$BA + \left[\frac{i((i-1)}{2} + j - 1\right] \times Size$$

Slide



Topic: Lower Triagular Matrix

A[1...100][1...100] is lower triangular matrix. Base address is 1000 and each element occupies 2-Bytes of space. What is the address of A[50][50] in row major order ?

a_{5050}

$$1000 + \left[\frac{50 \times 49}{2} + 49\right] \times 2$$

$$1000 + [25 \times 49 + 49] \times 2$$

$1000 + 2548$

3548

$1225 + 49 = 1274$

1274×2

2548

Slide



Topic: Lower Triagular Matrix

Before reaching 50th row, number of rows completed is 49

Total number of elements in 49 rows

1+2+3+.....+49 = (49×50)/2=1225

We are at 50th column , number of element completed is 49

Total number of element completed is 1225+49= 1274

Address of a[50][49] is = 1000+1274× 2=3548



2 mins Summary

Topic

Topic

Topic

Topic

Topic

Row major

$$BA + [(i - Lb_1) \times (uB_2 - LB_2 + 1) + (j - LB_2)] \times S$$

Column major - 2D

$$BA + [(j - Lb_2) \times (uB_1 - LB_1 + 1) + (i - Lb_1)] \times S_{ize}$$

LTM

Row major

$$BA + \left[\frac{i(i-1)}{2} + (j-1)\right] \times Size$$

$LB = 1$

Slide

THANK - YOU

Slide